

Performance Study of Serial Matrix Multiplication

Assignment #2, CSC 746, Fall 2026

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Abstract—TODO: In three or four sentences, state the focus of the study, the implementations and experimental approach, and the primary findings.

I. INTRODUCTION

TODO: Expand the abstract's problem statement into a short paragraph of three or four sentences.

TODO: Expand the approach statement into a short paragraph describing the three matrix-multiplication implementations and the performance study.

TODO: Expand the findings statement into a short paragraph summarizing the principal results.

II. IMPLEMENTATION

TODO: Briefly introduce the three implementations evaluated in this study. Use compact pseudocode where it improves clarity.

A. Basic Matrix Multiplication

TODO: Concisely describe the triply nested implementation of $C = C + AB$.

B. Blocked Matrix Multiplication with Copy Optimization

TODO: Describe the blocking/tiling implementation, the local copies of the A, B, and C blocks, the inner tile multiplication, and the supported block sizes.

C. CBLAS Matrix Multiplication

TODO: Describe the supplied CBLAS reference implementation and its role in correctness checking and performance comparison.

III. RESULTS

TODO: Briefly introduce the experiments and comparisons presented in this section. Report performance in MFLOP/s, with problem size on the horizontal axis for all charts.

A. Computational Platform and Software Environment

TODO: Describe the Perlmutter CPU hardware, cache and memory characteristics, operating environment, compiler version, and compilation flags.

B. Test Methodology

TODO: Explain what was measured and how. Include the timing boundaries, correctness check, problem sizes $N = \{128, 256, 512, 1024, 2048\}$, block sizes $B = \{2, 16, 32, 64\}$, FLOP count, MFLOP/s calculation, and treatment of the first CBLAS warm-up measurement.

C. Comparison of Basic Matrix Multiplication and CBLAS

TODO: Add a chart comparing Basic MM and CBLAS across all problem sizes. Discuss changes in Basic MM performance as problem size grows, explain those changes using course concepts, and compare Basic MM with CBLAS.

D. Blocked Matrix Multiplication with Copy Optimization and CBLAS

TODO: Add the five-variable chart containing CBLAS and the blocked implementation at $B = 2, 16, 32$, and 64 . Discuss effects of problem size and block size, compare the blocked implementation with CBLAS, and relate observed behavior to relevant hardware characteristics.

E. Overall Findings and Discussion

TODO: Compare the Basic and blocked-with-copy implementations. Explain where and why one outperforms the other, identify relevant hardware features, and discuss performance changes across the complete dataset and problem-size range.